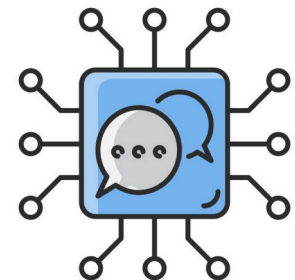


Speech Processing

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Agenda

- Articulatory Phonetics (Production)
- Acoustic Phonetics (Physics)
- Phonology (System)
- Computational Phonology
- DSP Techniques for Speech
- Automatic Speech Recognition (ASR)
- Text-to-Speech (TTS)
- Language Models

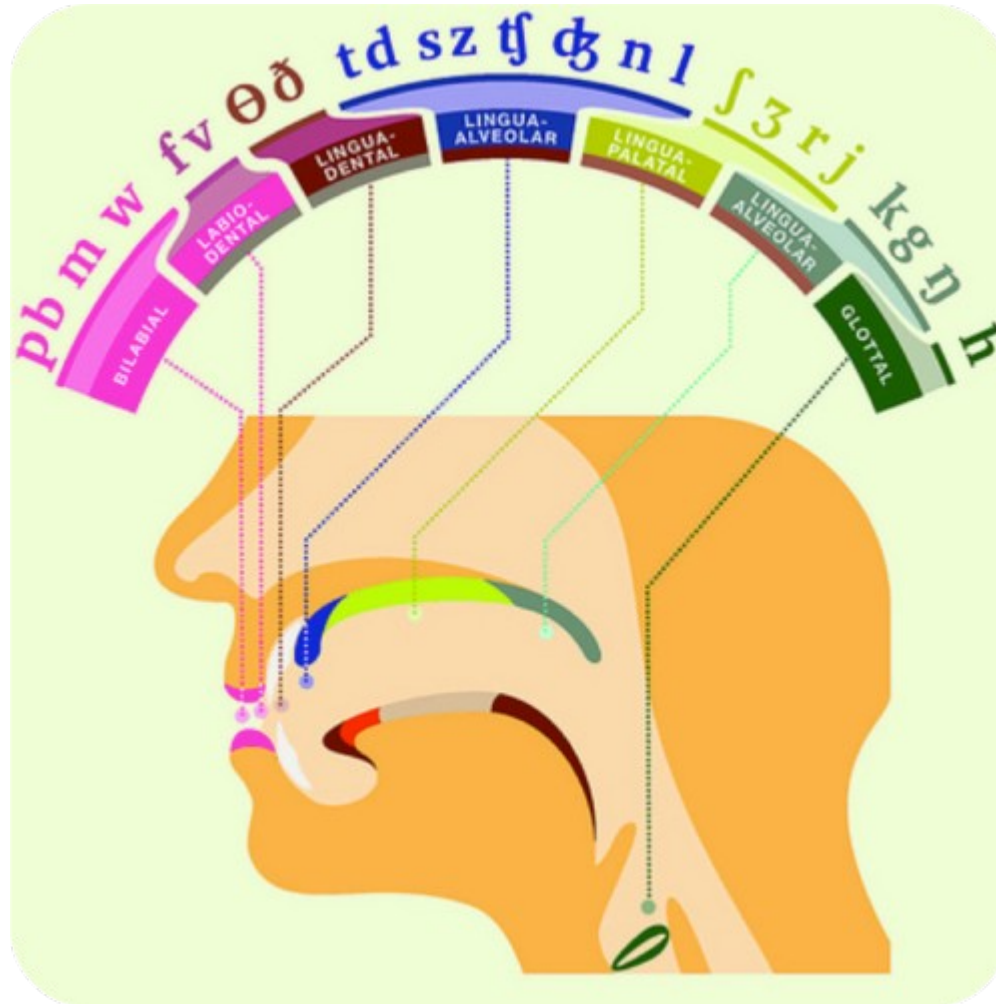
What is Speech?

- Speech is the most natural, efficient, and primary form of human communication.
- It is a continuous, time-varying acoustic signal carrying linguistic information (words, grammar) and paralinguistic information (emotion, gender, identity, health status).
- The Communication Chain: Thought (Psychological) → Articulation (Physiological) → Sound Wave (Physical) → Hearing (Physiological) → Understanding (Psychological).

Articulatory Phonetics

- Definition:
 - The branch of phonetics that studies how speech sounds are produced by the movement of the human vocal apparatus.
- Source-Filter Theory:
 - Speech is produced by a source (lungs/vocal cords) and modified by a filter (vocal tract).
- Key Mechanism:
 - Airflow from lungs (egressive pulmonic airstream) is obstructed or modified by articulators.

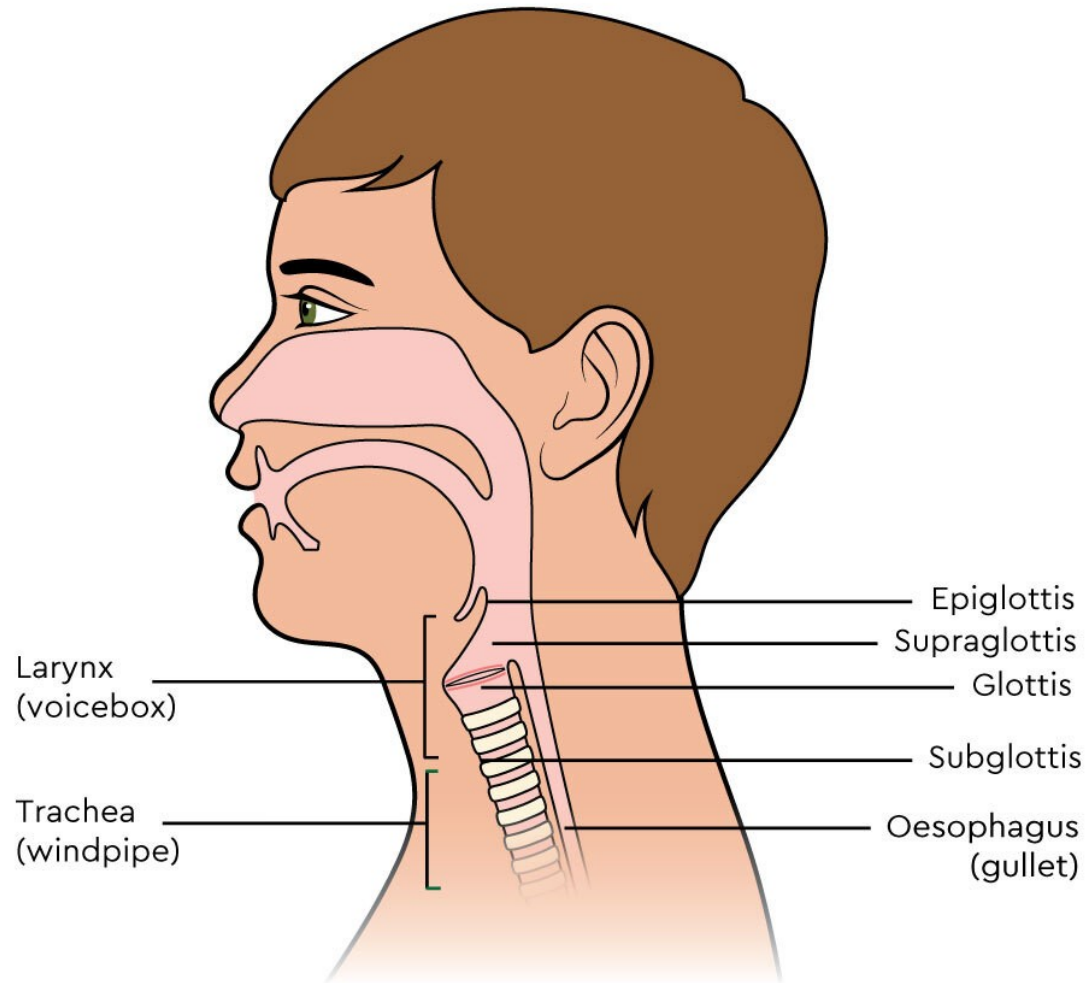
Articulatory Phonetics



The Vocal Cords (Phonation)

- Located in the Larynx (Voice Box).
- Voiced Sounds:
 - Vocal cords vibrate rapidly, chopping airflow into pulses. Examples: All vowels, /z/, /b/, /d/, /m/. (Feel your throat buzz).
- Voiceless Sounds:
 - Vocal cords are held open; air passes freely without vibration. Examples: /s/, /p/, /t/, /f/.
- Pitch (Fundamental Frequency F0):
 - Determined by the rate of vocal cord vibration. Higher tension = Higher Pitch.

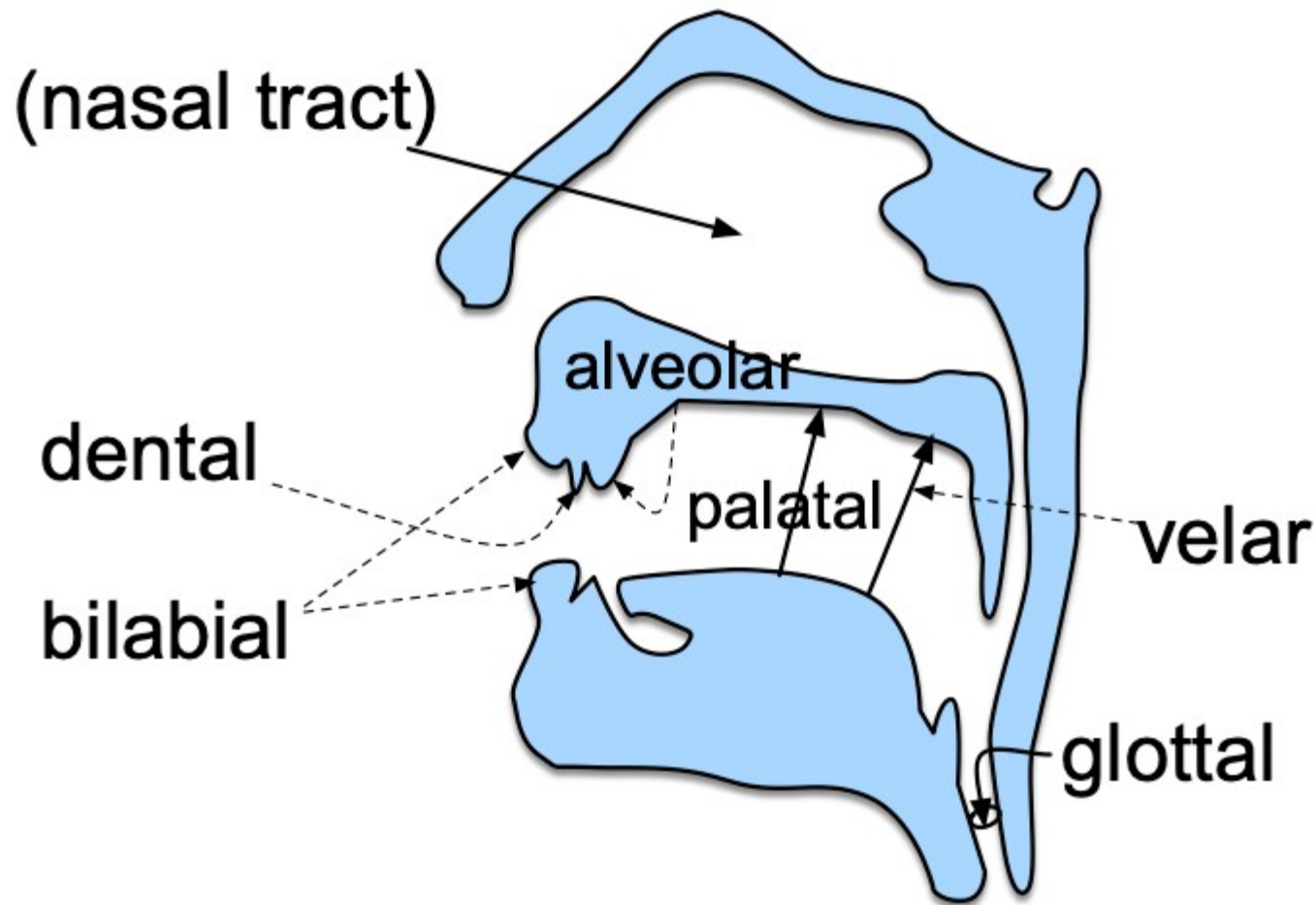
The Vocal Cords (Phonation)



Places of Articulation (Where?)

- Describes where the obstruction occurs in the vocal tract.
 - Bilabial: Both lips come together (/p/, /b/, /m/).
 - Labiodental: Lower lip touches upper teeth (/f/, /v/).
 - Alveolar: Tongue tip touches the alveolar ridge behind upper teeth (/t/, /d/, /s/, /z/, /n/).
 - Velar: Back of tongue touches the soft palate (velum) (/k/, /g/, /ŋ/).
 - Glottal: Constriction at the vocal cords (/h/).

Places of Articulation (Where?)



Manner of Articulation (How?)

- Describes how the airflow is modified.
- Stops (Plosives): Complete blockage followed by a sudden release (/p/, /k/, /t/).
- Fricatives: Narrow constriction causing turbulent airflow (hissing sound) (/f/, /s/, /sh/).
- Affricates: Begin as a stop, release as a fricative (e.g., "ch" in "chip", "j" in "jug").
- Nasals: Velum lowers, allowing airflow through the nose (/m/, /n/).
- Approximants (Liquids/Glides): minimal obstruction (/l/, /r/, /w/, /y/).
- Vowels: Open vocal tract; sound quality depends on tongue position and lip rounding.

Manner of Articulation (How?)

		PLACE							
MANNER		Bilabial	Labio-dental	Lingua-dental	Alveolar	Post-alveolar	Palatal	Velar	Glottal
Approximates	Stop	p, b			t, d			k, g	
	Fricative		f, v	θ, ð	s, z	ʃ, ʒ			h
	Affricate					tʃ, dʒ			
	Nasal	m			n			ŋ	
	Liquid (Lateral)				l				
	Liquid (Rhotic)				ɹ				
	Glide	w					j	w	

***Bolded** symbol is the voiced sound

Speech Sounds & Phonetic Transcription

- The Problem: English spelling (orthography) is inconsistent. "Fish" could theoretically be spelled "ghoti" (gh as in tough, o as in women, ti as in nation).
- The Solution: IPA (International Phonetic Alphabet). A standardized system where One Sound = One Symbol.
- Examples:
 - "Cat" → /kæt/
 - "Thought" → /θɔ:t/
 - "Judge" → /dʒʌdʒ/
- Essential for building pronunciation dictionaries in TTS and ASR.

Acoustic Phonetics

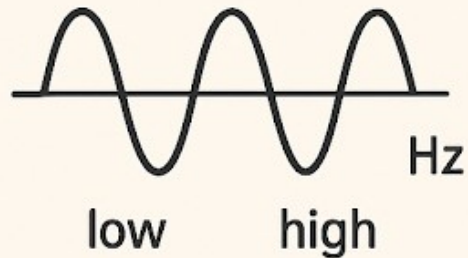
- Definition: The study of the physical properties of speech sound waves (frequency, amplitude, duration).
- Waveform: 2D plot of Amplitude vs. Time. Shows loudness variations and periodicity but hides frequency details.
- Spectrogram: 3D visualization (Time on X, Frequency on Y, Intensity/Amplitude as Color/Darkness). The most important tool for speech analysis.

Key Acoustic Features

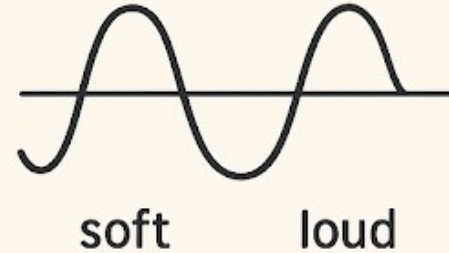
- Fundamental Frequency (F0): The lowest frequency of a periodic waveform; perceived as pitch.
- Harmonics: Integer multiples of F0.
- Formants (F1,F2,F3): Resonant frequencies of the vocal tract filter.
 - They appear as dark horizontal bands in a spectrogram.
 - Crucial for identifying Vowels.
 - F1: Correlates with tongue height (High tongue = Low F1).
 - F2: Correlates with tongue advancement (Front tongue = High F2).

Key Acoustic Features

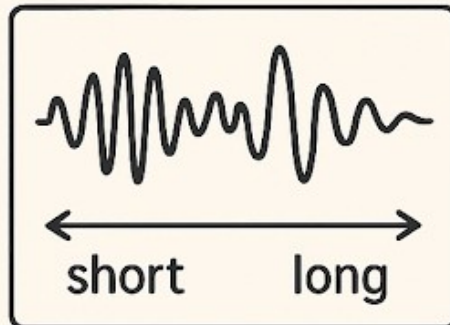
FREQUENCY



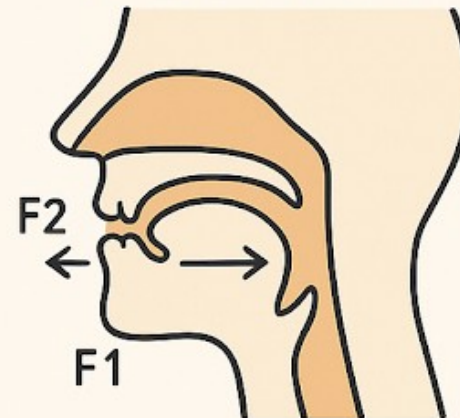
AMPLITUDE



DURAMTION



FORMANTS



Phonology (The System)

- Phonetics = Physical reality (continuous, infinite variations).
- Phonology = Abstract mental grammar (discrete categories).
- Phoneme: The smallest abstract unit of sound that distinguishes meaning in a language.
 - Minimal Pair Test: "pat" vs. "bat" → /p/ and /b/ are distinct phonemes.
- Allophone: Physical variations of a phoneme that do not change meaning.
 - Example: Aspirated ph in "pin" vs. unaspirated p in "spin". English speakers hear them as the "same" sound.

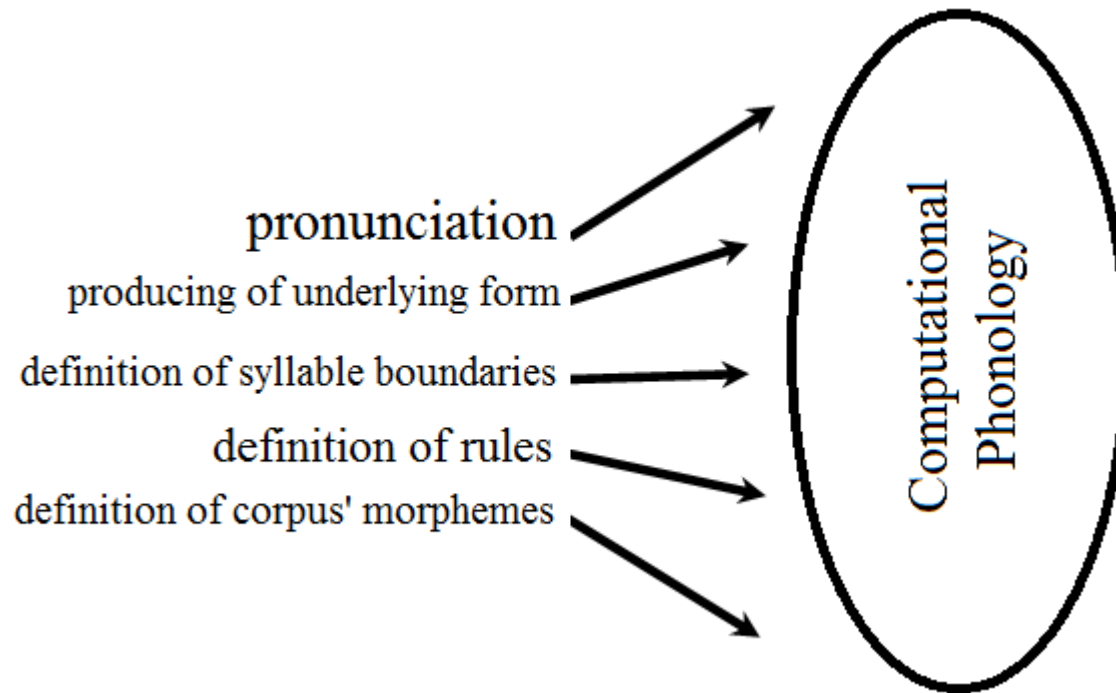
Phonology (The System)

Phonological feature	Sub-categories	Consonants
Place of articulation	Anterior	/b, m, f, θ, ð, ð ^ɹ , w/
	Medial	/l, t, d, d ^ɹ , t ^ɹ , n, r, s, z, s ^ɹ , ʃ, ʒ, j/
Manner of articulation	Posterior	/k, q, ʔ, χ, ɣ, ʕ, ʁ, h/
	Plosive	/b, d, t, k, q, ʔ, d ^ɹ , t ^ɹ /
	Fricative	/f, θ, ð, s, z, ʃ, χ, ʕ, s ^ɹ , ð ^ɹ , ɣ, ʁ, h/
	Affricative	/ʤ/
	Nasal	/m, n/
	Emphatic	/d ^ɹ , ð ^ɹ , s ^ɹ , t ^ɹ /
Voicing during articulation	Liquid and glide	/l, r, w, j/
	Voiced	/b, ʤ, d, ð, r, z, d ^ɹ , ð ^ɹ , ʁ, ɣ, l, m, n, w, j/
	Unvoiced	/ʔ, t, θ, χ, ʕ, s, ʃ, s ^ɹ , t ^ɹ , f, k, q, h/

Computational Phonology

- Goal: Modeling sound patterns and rules computationally.
- Grapheme-to-Phoneme (G2P): Converting written text (graphemes) to phonetic transcription. Critical for TTS.
 - Input: "Phone" → Output: /f oʊ n/
- Finite State Transducers (FSTs): The mathematical backbone for modeling phonological rules.
 - Example: Pluralization rule (s→z after voiced sounds).
 - Input: "dog" + "+s" → FST → Output: "dogz".

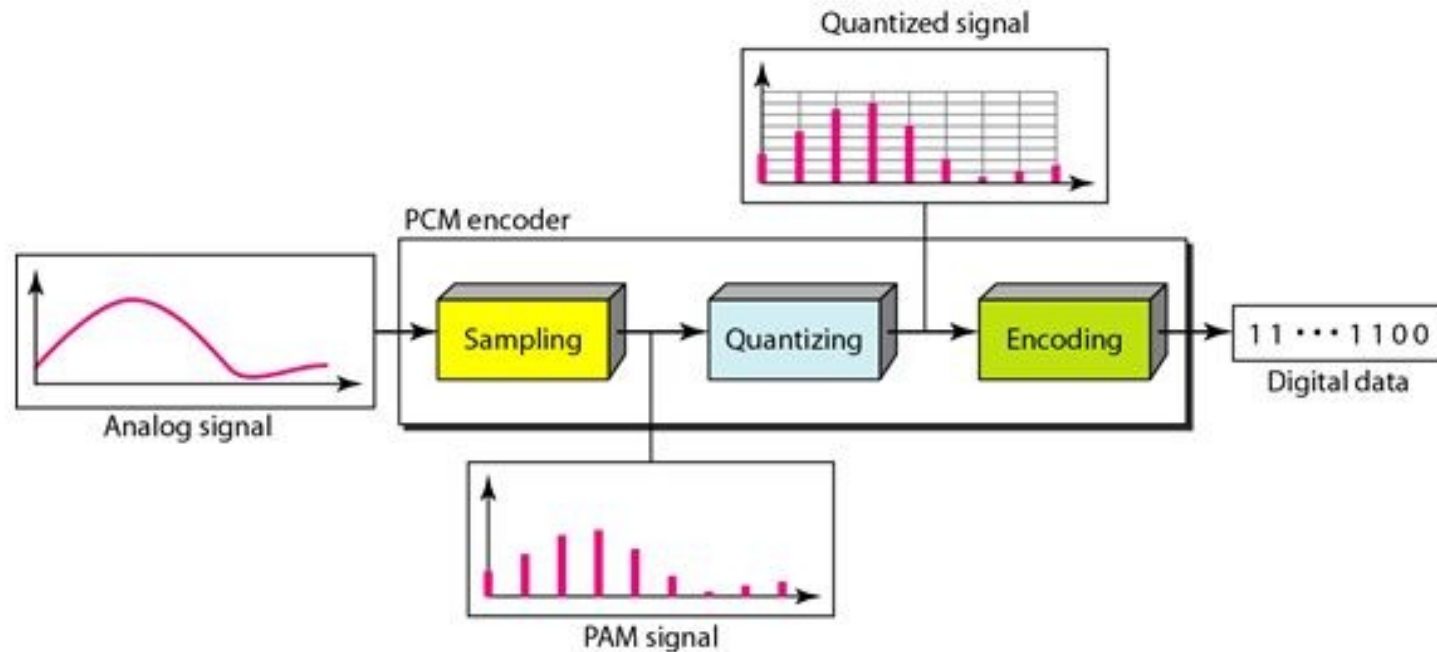
Computational Phonology



Analog to Digital Conversion

- Speech is naturally continuous (analog). Computers need discrete numbers (digital).
- Sampling: Measuring the wave amplitude at fixed time intervals.
 - Rate: 8kHz (Telephone), 16kHz (Wideband/ASR), 44.1kHz (CD Audio).
- Quantization: Rounding the amplitude value to the nearest integer (bit depth). e.g., 16-bit = 65,536 levels.
- Nyquist Theorem: To capture a frequency F , you must sample at least at $2F$. Human speech goes up to $\sim 8\text{kHz}$, so 16kHz sampling is standard.

Analog to Digital Conversion



DSP Techniques for Speech Pre-processing

- Pre-emphasis: Applying a high-pass filter to boost high frequencies (which naturally have lower energy). Makes formants clearer.
- Framing: Speech is non-stationary (changes constantly). We assume it is stationary for short periods (20-30ms). We slice the signal into overlapping frames.
- Windowing: Multiplying each frame by a window function (e.g., Hamming, Hanning) to taper the edges to zero, preventing spectral leakage (artifacts) during FFT.

Feature Extraction (MFCC)

- Mel-Frequency Cepstral Coefficients (MFCCs).
- The dominant feature for speech recognition before end-to-end Deep Learning.
- Philosophy: Mimic human ear perception. We hear pitch on a logarithmic scale (Mel Scale).
- Process:
 - Fast Fourier Transform (FFT) → Spectrum.
 - Apply Mel-spaced Filterbank (more filters in low frequencies).
 - Take Logarithm of energies (human hearing is log-compressed).
 - Discrete Cosine Transform (DCT) to decorrelate features.

What is ASR?

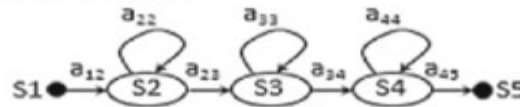
- Goal: To transcribe an audio waveform into a sequence of words (STT: Speech-to-Text).
- The Core Probabilistic Model: Find the word sequence W that maximizes probability given acoustic observation O .
- Bayes Rule: $P(W | O) = P(O) \cdot P(O | W) / P(W)$
 - $\text{argmax}(W_P(O | W) \cdot P(W))$
 - $P(O | W)$: Acoustic Model (How well the sound matches the phoneme).
 - $P(W)$: Language Model (How likely the word sequence is grammatically).

Traditional ASR Architecture (GMM-HMM)

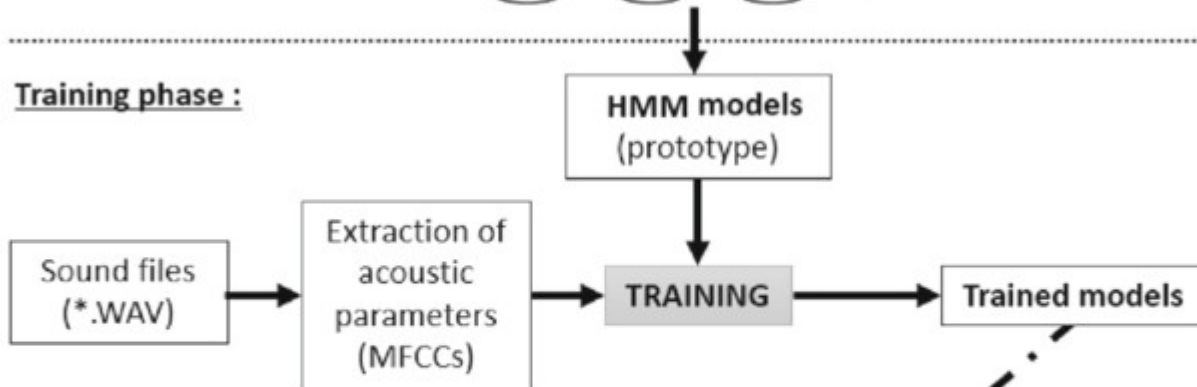
- Hidden Markov Model (HMM): Models the temporal evolution of speech. States represent phonemes or parts of phonemes. Transitions represent moving through time.
- Gaussian Mixture Model (GMM): Models the probability distribution of the spectral features (MFCCs) for each HMM state.
- Lexicon: A dictionary mapping Words → Phoneme sequences.
- Language Model: N-grams providing word probabilities.
- Decoding: Viterbi Algorithm finds the best path.

Traditional ASR Architecture (GMM-HMM)

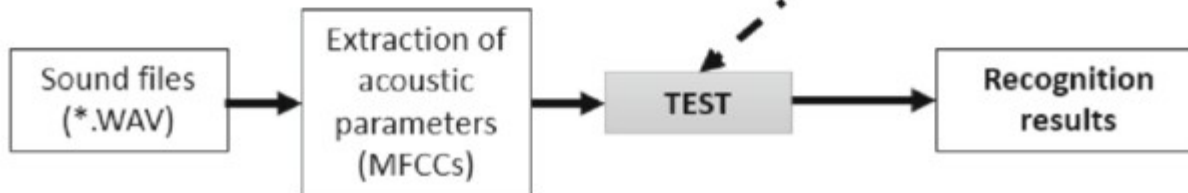
5 states HMM model of each phoneme :



Training phase :



Test phase :



Modern ASR Architecture

- Shift:
 - Deep Learning replaced the complex pipeline (GMMs/HMMs) with single neural networks.
- CTC (Connectionist Temporal Classification):
 - Loss function that solves the alignment problem (input is longer than output). Allows training RNNs directly on Audio-Text pairs.
- Seq2Seq with Attention (LAS - Listen, Attend, Spell):
 - Encoder-Decoder model. Encoder processes audio; Decoder generates characters, attending to specific audio frames.
- Transformers:
 - Wav2Vec 2.0 (Self-supervised learning on raw audio), Whisper (OpenAI - trained on massive internet data).

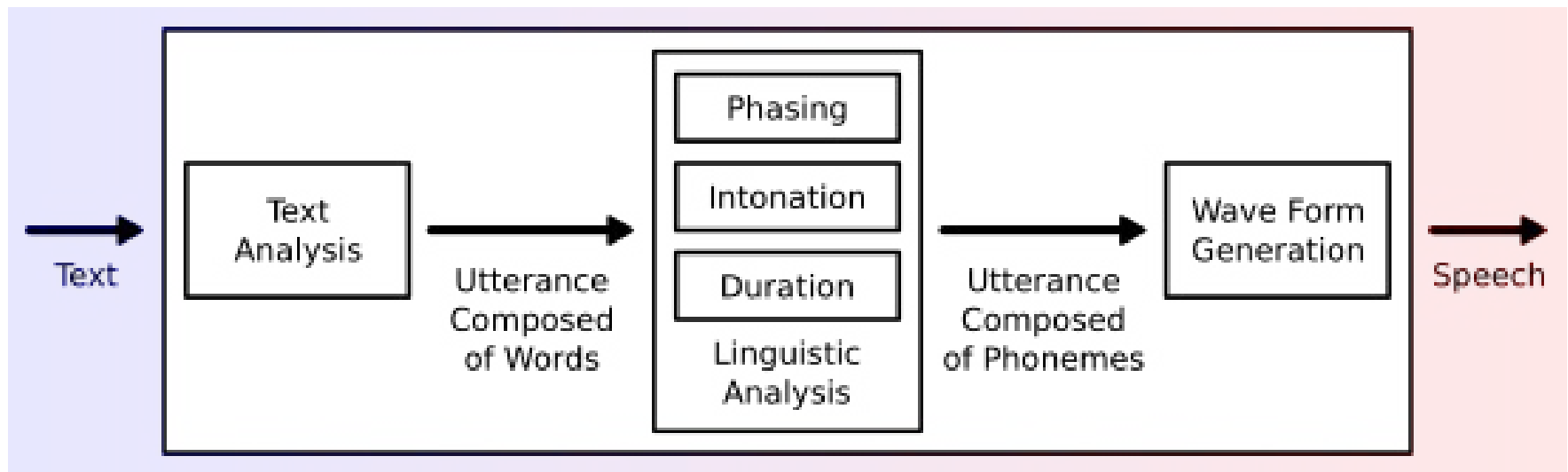
Challenges in ASR

- Noise Robustness: "Cocktail Party Problem" (separating voice from background noise/other speakers). Reverb in large rooms.
- Speaker Variability: Accents, dialects, speed of speech, age, gender, emotional state.
- Vocabulary: Out-of-vocabulary (OOV) words like names or new slang.
- Homophones: "To", "Two", "Too" sound identical. ASR relies heavily on the Language Model to pick the right one based on context.
- Code-Switching: Mixing languages in one sentence (e.g., Hinglish).

What is TTS?

- Goal: Generate a natural-sounding, intelligible speech waveform from input text.
- Also called: Speech Synthesis.
- Pipeline:
 - Text Analysis (Frontend): Normalization ("Dr." → "Doctor", "\$10" → "Ten dollars", "1998" → "Nineteen ninety-eight").
 - Linguistic Analysis: Grapheme-to-Phoneme (G2P), Prosody prediction (stress, intonation).
 - Waveform Generation (Backend): Creating the sound.

What is TTS?



Speech Synthesis Approaches - 1: Concatenative

- Method: "Cut and Paste" approach.
- Process:
 - Record a voice actor reading thousands of sentences. Slice the audio into tiny units (diphones: middle of one phone to middle of next).
- Synthesis:
 - Select best units from database and glue them together.
- Pros:
 - Highly natural sound quality (actual human voice).
- Cons:
 - "Glitchy" transitions; Cannot change voice style, emotion, or speed easily; Requires huge storage.

Speech Synthesis Approaches - 2: Parametric

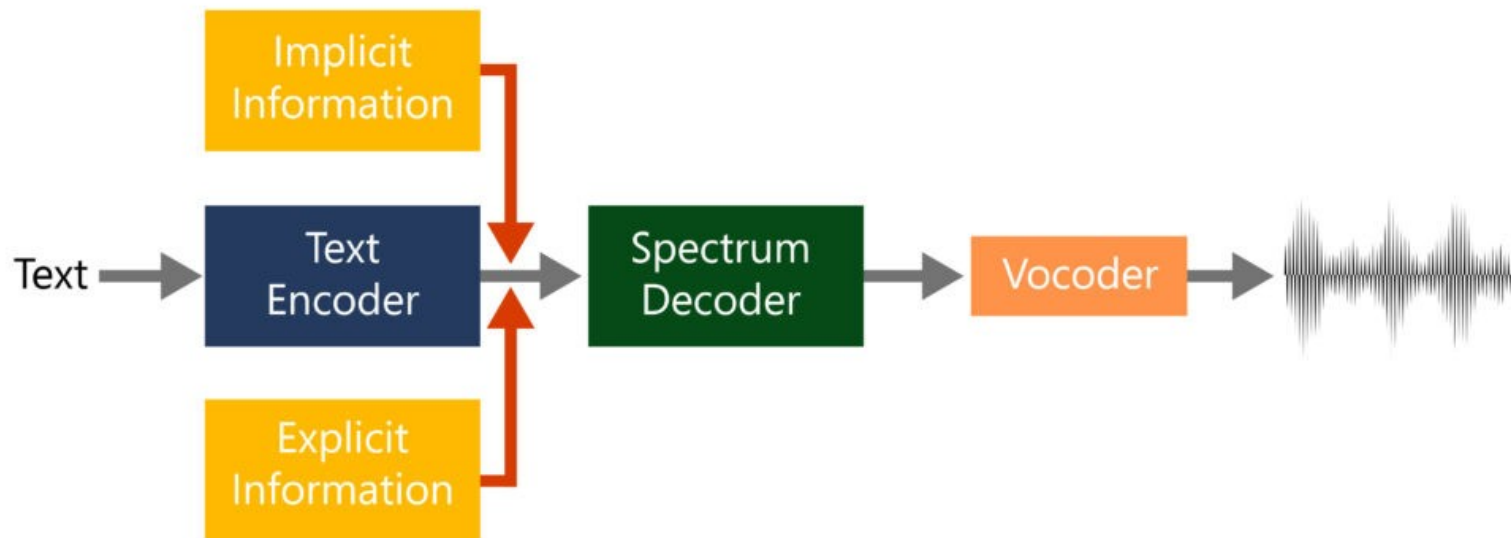
- Method: Statistical generation.
- Process:
 - Don't store audio. Instead, learn statistical models (HMMs) of parameters like pitch (F0), duration, and spectral envelope.
- Synthesis:
 - Generate parameter trajectories and pass them through a Vocoder to create sound.
- Pros:
 - Flexible (can change emotion/speaker easily), tiny footprint.
- Cons:
 - Sounds "muffled," "buzzy," or robotic. Lacks the crispness of real speech.

Speech Synthesis Approaches - 3: Neural TTS (Modern)

- Method: Deep Learning End-to-End.
- Acoustic Model (e.g., Tacotron 2, FastSpeech):
 - Sequence-to-Sequence model mapping Text/Phonemes directly to Mel Spectrograms.
- Neural Vocoder (e.g., WaveNet, HiFi-GAN):
 - Generative model converting Mel Spectrograms to high-fidelity raw audio waveforms sample-by-sample.
- Result:
 - Near-human quality. Solved the "robotic" sound issue.

Neural TTS Pipeline

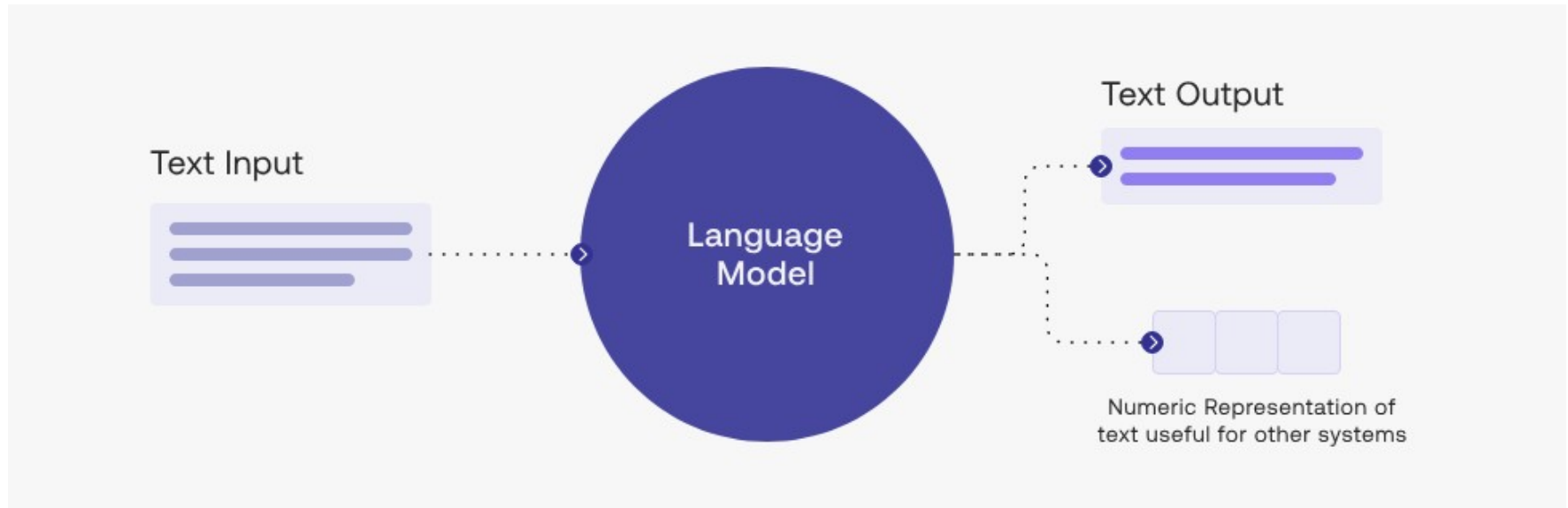
Azure Neural TTS model Uni-TTSv4 shows no significant difference in quality from sentence-level natural speech recordings



What is a Language Model (LM)?

- Definition: A statistical tool that assigns a probability to a sequence of words.
- Function: It predicts the next word in a sequence.
 - $P(\text{"I am going to"}) \rightarrow \text{"school"}(\text{High}), \text{"banana"}(\text{Low})$.
- Crucial Role:
 - ASR: Helps distinguish between "recognize speech" and "wreck a nice beach".
 - Translation: Ensures output is grammatically correct.
 - Autocomplete: Suggests next words.

What is a Language Model (LM)?



N-Gram Language Models

- Markov Assumption: The probability of the next word depends only on the previous $N-1$ words.
 - Unigram ($N=1$): Probability based on word frequency alone. No context.
 - Bigram ($N=2$): Look at 1 previous word. $P(\text{word}_i | \text{word}_{i-1})$.
 - Trigram ($N=3$): Look at 2 previous words.
- Sparsity Problem: If a 3-word sequence never appeared in training data, probability is 0. (Solved by Smoothing techniques).

Evaluation of Language Models

- Perplexity (PP): The standard metric.
- Concept:
 - How "surprised" or "confused" is the model by real test data?
- Calculation:
 - Inverse probability of the test set, normalized by the number of words.
- Rule:
 - Lower Perplexity = Better Model. A perfect model would have perplexity near 1.

Neural Language Models

- RNN / LSTM LMs: Unlike N-grams, they have "infinite" context history via hidden states. They learn distributed representations (embeddings).
- Transformer LMs:
 - Masked LM (BERT): Predict missing words in the middle. Good for understanding.
 - Causal LM (GPT): Predict the next word. Good for generation.
- Advantage: Better generalization. "Cat sat on mat" helps predict "Dog sat on rug" due to semantic similarity in vector space.

Indian Language Script Technology

- Unique Challenges:
 - Complex Scripts: Abugida writing systems (Consonant-Vowel combinations).
 - Agglutinative Morphology: Words are formed by stringing morphemes together (Dravidian languages).
 - Low Resource: Less digital data available compared to English.
- Tools:
 - Unicode: Standard encoding for all Indic scripts.
 - Transliteration: Converting script (e.g., Marathi to Roman) for easier input.
 - Projects: AI4Bharat (IndicWav2Vec, IndicBART) creating open-source datasets and models.

Summary

- Speech processing sits at the intersection of Linguistics, Physics (Acoustics), and Computer Science (AI).
- ASR is the "Ear" (Bottom-up processing from signal to symbol).
- TTS is the "Mouth" (Top-down processing from symbol to signal).
- Deep Learning (Transformers) has revolutionized both, enabling systems like Siri, Alexa, and Google Translate to achieve near-human performance.

Thank you

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